

Erdős Problem #160

EP-160

L1 - TRACTABLE

OPEN

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $h(N)$ be the smallest k such that $\{1, \dots, N\}$ can be coloured with k colours so that every four-term arithmetic progression must contain at least three distinct colours. Estimate $h(N)$.

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>